



Enfermedades Infecciosas y Microbiología Clínica

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Consensus document

Executive summary of the consensus document of the Spanish Society of Infectious Diseases and Clinical Microbiology (SEIMC: GEPI, GeSIDA, GESITRA-IC, GEIRAS) on screening for imported infectious diseases in immunocompromised patients



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ABSTRACT

The increase in global mobility has led to a higher prevalence of imported diseases in immunocompromised patients, often asymptomatic but with the potential for reactivation or severe progression. This executive summary presents the key recommendations of the consensus document developed by the Imported Pathology Study Group (GEPI-SEIMC) in collaboration with GeSIDA, GESITRA and GEIRAS (SEIMC), targeting healthcare professionals who manage immunocompromised individuals. Based on a structured narrative review, the document proposes systematic screening strategies to detect imported infections during their asymptomatic phase, clearly distinguishing them from the clinical approach in symptomatic patients. Major infections of concern are summarized. Specific guidance is provided for people living with HIV, donors and transplant recipients, oncohematologic patients, and those receiving immunosuppressive therapy. Differences in reactivation risk, preferred diagnostic methods, and therapeutic decisions in cases of latent infection are highlighted. The document underscores the need to incorporate screening into pre-treatment and pre-transplant assessments, promoting standardized protocols.

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Documento de consenso de la Sociedad Española de Enfermedades Infecciosas y Microbiología Clínica (SEIMC: GEPI, GESITRA-IC, GeSIDA, GEIRAS) sobre cribado de enfermedades infecciosas importadas en pacientes inmunodeprimidos

RESUMEN

El aumento de la movilidad global ha dado lugar a una mayor prevalencia de enfermedades importadas en pacientes inmunodeprimidos, a menudo asintomáticas pero con potencial de reactivación o evolución grave. Este resumen ejecutivo presenta las recomendaciones clave del documento de consenso elaborado por el Grupo de Estudio de Patología Importada (GEPI-SEIMC), en colaboración con GeSIDA, GESITRA y GEIRAS (SEIMC), dirigido a profesionales sanitarios que atienden a personas inmunodeprimidas. Basado en una revisión narrativa estructurada, el documento propone estrategias de cribado sistemático para detectar infecciones importadas en fase asintomática, diferenciándolas claramente del abordaje clínico en pacientes sintomáticos. Se resumen las principales infecciones de interés. Se ofrecen orientaciones específicas para personas que viven con el VIH, donantes y receptores de trasplante, pacientes oncohematológicos y personas en tratamiento inmunosupresor. Se destacan las diferencias en cuanto al riesgo de reactivación, métodos diagnósticos preferentes y decisiones terapéuticas en casos de infección latente. El documento subraya la necesidad de incorporar el cribado en las evaluaciones previas al tratamiento y al trasplante, promoviendo protocolos estandarizados.

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Introduction

Increased global mobility has led to a higher incidence of imported diseases in non-endemic areas. In immunosuppressed individuals, some pathogens may reactivate or progress severely or disseminate. Therefore, it is essential to implement screening in patients with risk factors – distinct from clinical diagnosis, which targets individuals with symptoms.

Risk assessment must consider the country of origin, migration history, pathogen exposure, travel to tropical areas, and patient characteristics.

A narrative review was conducted for the development of this document, incorporating the GRADE (Grading of Recommendations Assessment, Development and Evaluation) system to formulate recommendations. GRADE provides a transparent and systematic approach to evaluating the quality of evidence and the strength of recommendations. Recommendations are classified as either

strong or weak, based on the quality of available evidence, which may be rated as high, moderate, low, or very low.

Given the continuous evolving prevalence and epidemiology of many imported infections, there is a possibility that some conditions currently considered emerging or re-emerging may not have been included in this document but could become increasingly relevant in the future. Therefore, it is recommended to remain alert to international health advisories and shifts in global epidemiological trends, and to consult with experts as needed.

Which pathogens require screening and where are they found?

1. *Chagas disease* caused by *Trypanosoma cruzi*, it is endemic in 21 Latin American countries, with vector transmission in rural areas. Migration has enabled its spread to other continents; in

Spain, an estimated 50,000 cases exist, though only 30% are diagnosed. Beyond vector transmission, autochthonous cases have been reported via transfusions, transplants, and vertical transmission. In immunosuppressed patients, it may cause CNS infection (chagoma) and reactivation syndromes. *Recommendation*: Screen patients from Latin America (except the Caribbean) (strong recommendation; moderate evidence), women of child-bearing age and/or pregnant (strong recommendation; high evidence), and children of mothers from endemic areas (strong recommendation; high evidence).

2. *Malaria* caused by five *Plasmodium* species, with *P. falciparum* being the most lethal. It is endemic in 83 countries: Sub-Saharan Africa (Nigeria, Democratic Republic of Congo, Tanzania, and Mozambique account for >50% of deaths), India in Asia, Papua New Guinea in the Western Pacific, and Venezuela and Brazil in the Americas. Submicroscopic malaria has been described up to 3 years after leaving endemic areas. *Recommendation*: Screen patients who stayed in endemic areas in the last 3 years, especially Sub-Saharan Africa (strong recommendation; low evidence).
3. *Strongyloidiasis* caused by *Strongyloides stercoralis*, globally distributed but more prevalent in tropical and humid regions. It is often asymptomatic in immunocompetent individuals but may progress to hyperinfection or dissemination – especially in HTLV-1 co-infected patients – with high mortality. Prevalence in Spain among migrants ranges from 8 to 20%. The parasite can live for over 40 years in humans. *Recommendation*: Screen patients from endemic areas regardless of time since arrival (strong recommendation; moderate evidence).
4. *Schistosomiasis* caused by *Schistosoma* spp. (*S. mansoni*, *S. haematobium*, *S. japonicum*), acquired through contact with contaminated freshwater. It is endemic in Sub-Saharan Africa, the Middle East, Asia, and South America. It may lead to hepatic fibrosis, portal hypertension, genitourinary or neurological involvement, more severely in immunocompromised individuals. *Recommendation*: Screen patients from endemic areas regardless of time since arrival (strong recommendation; moderate evidence).
5. *Intestinal parasitosis and soil-transmitted helminths* These include protozoan and helminth infections, with a global prevalence of ~25%, especially in children aged 1–14: *Ascaris lumbricoides* (~804 million infected), *Trichuris trichiura* (~477 million), *Ancylostoma duodenale* and *Necator americanus* (~472 million). Immunosuppression significantly increases the risk of severe disease regardless of geographical endemicity. *Recommendation*: Screen patients from any endemic area who arrived within the last 3 years (weak recommendation; moderate evidence).
6. *Endemic mycoses* caused by dimorphic fungi endemic to specific tropical and subtropical regions that convert into yeasts upon inhalation. They affect both immunocompetent and immunocompromised individuals. Key agents include *Histoplasma capsulatum* (cosmopolitan with foci in the Americas, Africa, China, and some autochthonous cases in Europe), *Coccidioides* and *Paracoccidioides* (Americas), *Blastomyces dermatitidis*, and *Emergomyces* spp. *Recommendation*: Monitor patients from endemic areas without specific screening (strong recommendation; very low evidence).
7. *Tuberculosis (TB)* caused by *Mycobacterium tuberculosis*, with high incidence in Sub-Saharan Africa, the Indian subcontinent, Southeast Asia, and China. HIV coinfection exceeds 50% in southern Africa; multidrug-resistant TB (MDR/RR-TB) accounts for 50% of cases in former Soviet states. *Recommendation*: Screen patients from any endemic area regardless of time since arrival (strong recommendation; high evidence).
8. *Arboviruses* Arthropod-borne viral diseases that are emerging due to vector expansion, environmental change, and human

mobility. In Spain, autochthonous cases include Toscana virus (VTOS), West Nile virus (WNV), and Crimean-Congo hemorrhagic fever virus (CCHFV). Imported viruses of public health concern include dengue (DENV), chikungunya (CHIKV), Zika (ZIKV), Mayaro, Rift Valley fever, and Japanese encephalitis. *Recommendation*: No systematic screening. Screen blood donors and transplant candidates in the presence of epidemiological risk or local WNV outbreaks (strong recommendation; very low evidence).

What screening techniques should be used?

1. *Chagas disease*: Screening should involve two parallel serological tests based on different principles, or one highly sensitive test followed by confirmation with a highly specific test based on a different principle (strong recommendation; moderate evidence).
2. *Malaria*: High-sensitivity molecular techniques such as PCR are recommended for screening asymptomatic individuals (strong recommendation; high evidence).
3. *Strongyloidiasis*: IgG ELISA serology is recommended in immunocompromised patients (strong recommendation; low evidence); empirical treatment may be considered if clinical suspicion is high and diagnostic tests are unavailable.
4. *Schistosomiasis*: Serological screening is recommended for all individuals from endemic areas (strong recommendation; moderate evidence).
5. *Intestinal parasitosis and soil-transmitted helminths*: Microscopic examination of three concentrated stool samples is advised.
6. *Endemic mycoses*: Routine screening is not recommended due to diagnostic limitations. In selected cases, serology and molecular methods may be combined.
7. *Tuberculosis*: Screening with IGRA is recommended (strong recommendation; moderate evidence).
8. *Arboviruses*: Acute phase (<7 days): direct diagnosis through viral isolation, NS1 antigen detection (dengue), or RT-PCR. Convalescent phase: serology (IgM, IgG, neutralizing antibodies); interpretation should consider cross-reactivity and variability in immunocompromised individuals or pregnant women. Systematic screening is not recommended (strong recommendation; very low evidence).

How should specimens be managed and results communicated?

Proper management of clinical specimens and effective communication between laboratory and clinical teams are critical for diagnosing imported infections in immunocompromised patients, especially transplant recipients. These individuals are at increased risk for severe complications from pathogens uncommon in non-endemic areas and require specialized care.

The formation of multidisciplinary teams – including infectious disease specialists, microbiologists, and pharmacists – is essential. These teams should create and regularly update protocols tailored to local resources and patient populations, ensuring diagnostic accuracy and patient safety.

Structured, documented communication is strongly encouraged. While verbal interactions are useful, digital health systems enhance traceability, reduce errors, and accelerate clinical decision-making. These systems can support predefined test requests, sample collection guidelines, and automated alerts for critical results.

Turnaround times should be agreed upon by all involved professionals, based on clinical urgency. Every phase – from ordering

to result delivery – should be monitored to identify delays and improve efficiency.

Sample handling must follow all current regulations for collection, storage, and transport. Accurate interpretation relies on detailed clinical information, including geographic origin, travel history, type and timing of immunosuppression, and recent blood product administration.

Finally, potentially hazardous specimens require appropriate biosafety measures and mandatory disease reporting. Retaining pre-immunosuppression samples may be useful for retrospective testing in selected cases provided ethical and biobanking standards are strictly followed.

Recommendations

1. *Establish multidisciplinary teams* tailored to local settings and available resources to lead the design, implementation, and dissemination of diagnostic protocols (*strong recommendation; very low level of evidence*).
2. *Use digital tools to improve diagnostic accuracy*: Implement pre-defined test request profiles, automated Information Technology rules, and alert systems to support diagnostic quality and timely response (*strong recommendation; very low level of evidence*).
3. *Ensure accessible information and rapid communication*: Maintain clear and accessible instructions for specimen collection and guarantee urgent communication of critical results, particularly in high-risk contexts such as transplantation (*strong recommendation; very low level of evidence*).

Which people living with HIV (PLWH) require screening?

Recommendations

1. All migrant PLWH (*strong recommendation; moderate evidence*).
2. PLWH with severe immunosuppression ($CD4^+ < 200$ cells/ μ L) from endemic areas (*strong recommendation; high evidence*).
3. PLWH with $CD4^+ < 100$ cells/ μ L:
 - a. Screen for cryptococcosis using cryptococcal antigen (CrAg) detection in serum, plasma, or whole blood (*strong recommendation; moderate evidence*). If positive, perform lumbar puncture to rule out CNS involvement.
 - b. Screen for *Cryptosporidium* spp. and *Cystoisospora* spp. in stool (*strong recommendation; moderate evidence*).
4. Women living with HIV of childbearing age and/or pregnant women, and children born to mothers from endemic areas (*strong recommendation; high evidence*) in *T. cruzi* screening.
5. PLWH who are migrant travelers visiting friends and relatives (VFR) are at elevated risk for travel-related diseases and may benefit from post-travel screening based on risks, even if asymptomatic (*weak recommendation; low evidence*).
6. PLWH undertaking long-term travel (>3 months) to high-risk areas or engaging in specific activities may also benefit from screening. Repeat screening for diseases that do not confer permanent immunity should be considered based on destination and activities (*strong recommendation; high evidence*).
7. Systematic screening is not recommended for PLWH with well-controlled immunovirological status who travel internationally, unless they present clinical signs or symptoms after travel (*strong recommendation; high evidence*).

Which transplant patients require screening?

Please refer to the GESITRA-ONT document: <https://www.ont.es/wp-content/uploads/2023/06/CRITER2.pdf> or its updates.

Recommendations

- (a) Donors (solid organ or hematopoietic stem cell donors) (*strong recommendation; low level of evidence*)
 1. *From Central and South America*: Screen for *Trypanosoma cruzi* (except Caribbean islands), *Plasmodium* spp. (especially in Central America and the Amazon), *Strongyloides stercoralis*, *Schistosoma* spp. (only in the Caribbean, Venezuela, and Brazil); *Histoplasma capsulatum* (serology), *Coccidioides immitis*, arboviruses, and HTLV-1. Include individuals who have received blood transfusions in risk areas or are children of mothers from endemic regions in *T. cruzi* screening.
 2. *From North Africa*: Screen for *Strongyloides stercoralis*, *Schistosoma* spp., and HTLV-1.
 3. *From Sub-Saharan Africa*: Screen for *Plasmodium* spp., *Strongyloides stercoralis*, *Schistosoma* spp., *Histoplasma capsulatum* (in West Africa, using serology), arboviruses, and HTLV-1.
 4. *From the Indian subcontinent*: Screen for *Plasmodium* spp., *Strongyloides stercoralis*, and HTLV-1.
 5. *From Southeast Asia*: Screen for *Plasmodium* spp., *Strongyloides stercoralis*, *Schistosoma* spp., *Clonorchis/Opisthorchis* spp. (only in liver donors), arboviruses, and HTLV-1.
 6. It is not necessary to screen hematopoietic stem cell donors for *Strongyloides stercoralis*, *Schistosoma* spp., *Histoplasma capsulatum*, *Coccidioides immitis*, or *Clonorchis/Opisthorchis* spp.
- (b) Recipients (solid organ or hematopoietic stem cell transplant recipients) (*strong recommendation; low level of evidence*)
 1. *From Central and South America*: Screen for *Trypanosoma cruzi* (except Caribbean islands), *Plasmodium* spp. (especially in Central America and the Amazon), *Strongyloides stercoralis*, *Schistosoma* spp. (only in the Caribbean, Venezuela, and Brazil); *Histoplasma capsulatum* (serology), *Coccidioides immitis*, arboviruses, and HTLV-1. Include individuals who have received blood transfusions in risk areas or are children of mothers from endemic regions in *T. cruzi* screening.
 2. *From North Africa*: Screen for *Strongyloides stercoralis*, *Schistosoma* spp., and HTLV-1.
 3. *From Sub-Saharan Africa*: Screen for *Plasmodium* spp., *Strongyloides stercoralis*, *Schistosoma* spp., *Histoplasma capsulatum* (West Africa, using serology), and HTLV-1.
 4. *From the Indian subcontinent*: Screen for *Plasmodium* spp., *Strongyloides stercoralis*, and HTLV-1.
 5. *From Southeast Asia*: Screen for *Plasmodium* spp., *Strongyloides stercoralis*, *Schistosoma* spp., and HTLV-1.
 - **Tuberculosis – Infection by *Mycobacterium tuberculosis* (MTI)**: In living donors, tuberculin skin testing and/or IGRA should be performed. If positive, active TB must be ruled out before transplantation. In deceased donors, evaluation should be individualized based on clinical history, epidemiology (residence in high-incidence areas), and radiological findings. MTI screening should also be performed in all solid organ and hematopoietic stem cell recipients.

Which patients undergoing chemotherapy require screening?

Evidence on the reactivation of imported infections in this context is scarce. It is necessary to assess the drug's profile, intrinsic immunodeficiency, number and type of prior therapeutic lines, and cumulative exposure to corticosteroids or purine analogs (fludarabine, cladribine, thioguanine).

Recommendations

1. *In recipients of allogeneic hematopoietic stem cell transplantation (allo-HSCT) and patients undergoing CAR-T cell therapy:* there is only evidence of risk of tuberculosis infection reactivation, which should be screened (strong recommendation; low level of evidence).
2. *Among biological agents,* Janus kinase (JAK) inhibitors such as ruxolitinib (selective for JAK-1 and JAK-2), Bruton's tyrosine kinase (BTK) inhibitors (ibrutinib), and phosphatidylinositol 3-kinase δ (PI3K δ) inhibitors (idelalisib) require MTI screening and treatment (strong recommendation; low level of evidence). There is a theoretical but unconfirmed risk of reactivating latent histoplasmosis with these drugs.
3. *Immune checkpoint inhibitors* (anti-PD-1/PD-L1 and anti-CTLA-4 antibodies) are not immunosuppressants per se but often require high-dose corticosteroids or anti-TNF- α agents to manage immune-related adverse events (irAEs). This implies a risk of unmasking or reactivating tuberculosis and strongyloidiasis. MTI and *Strongyloides* screening is recommended before initiating checkpoint inhibitors (strong recommendation; low level of evidence).
4. *Anti-CD20 antibody treatment (rituximab):* Screening for *S. stercoralis* is recommended, especially if corticosteroids are used (weak recommendation; very low level of evidence).

Which patients starting immunosuppressive therapy require screening?

Recommendations by pharmacological class

1. *TNF antagonists (anti-TNF):* Infliximab, adalimumab, golimumab, certolizumab pegol, and etanercept are associated with reactivation or dissemination of endemic parasitic and fungal infections. There is evidence of reactivation of *Strongyloides stercoralis* and *Trypanosoma cruzi* in patients from endemic areas, particularly when corticosteroids are used concomitantly. Tuberculosis reactivation is especially observed in rheumatoid arthritis. *Recommendations* – Screening is recommended for Chagas disease and strongyloidiasis (strong recommendation; low level of evidence), and for *M. tuberculosis* infection (MTI) before initiating treatment (strong recommendation; high level of evidence). Systematic screening for endemic mycoses (histoplasmosis, coccidioidomycosis) is not justified (weak recommendation; very low level of evidence).
2. *IL-6 antagonists:* Tocilizumab, sarilumab, and siltuximab share the risk of TB reactivation with anti-TNF agents. *Recommendations* – MTI screening is recommended prior to administration (strong recommendation; moderate level of evidence).
3. *IgE and IL-5 antagonists:* Omalizumab and anti-IL-5 agents (benralizumab, mepolizumab, reslizumab) may impair defenses against helminths. Clinical trials show a trend toward increased helminth infections (*Ascaris*, *Trichuris*). *Recommendations* – Screening for strongyloidiasis and soil-transmitted helminths is recommended in patients from endemic areas (strong recommendation; moderate level of evidence).
4. *JAK-STAT inhibitors:* Tofacitinib, ruxolitinib, momelotinib, among others, pose risks for fungal infections (e.g., cryptococcosis) and TB, especially at high doses of tofacitinib in patients from endemic regions. *Recommendations* – Systematic screening for endemic mycoses is not recommended (weak recommendation; very low level of evidence). MTI screening is advised in migrants from regions with moderate-to-high TB incidence (strong recommendation; moderate level of evidence).

5. *Drugs for multiple sclerosis and neuromyelitis optica:* Dimethyl fumarate, fingolimod, alemtuzumab, cladribine, and teriflunomide have been associated with TB and fungal infections. They may also interfere with MTI testing. *Recommendations* – MTI screening is recommended before starting treatment (strong recommendation; low level of evidence), and clinical evaluation with chest X-ray is advised before initiating fingolimod, alemtuzumab, or natalizumab to rule out fungal infections (weak recommendation; very low level of evidence).

What treatment should be applied?

1. *Chronic Chagas disease:* Treatment with benznidazole or nifurtimox is recommended before initiating immunosuppression, although its efficacy in preventing reactivation is not well established. Follow-up should include clinical surveillance and, for untreated cases, PCR monitoring during immunosuppression (strong recommendation; moderate level of evidence).
2. *Malaria:* Treatment is indicated in all confirmed cases, preferably using artemisinin-based regimens. For infections caused by *P. vivax* or *P. ovale*, primaquine should be added after ruling out G6PD deficiency. Clinical and parasitological follow-up is essential to confirm cure (strong recommendation; high level of evidence).
3. *Strongyloidiasis:* Treatment with ivermectin is recommended, including empirically before immunosuppression if timely exclusion of infection is not possible. Follow-up should include serology, stool studies, and evaluation of eosinophilia. In *Loa loa* endemic areas, microfilaria screening is necessary prior to ivermectin administration (strong recommendation; moderate level of evidence).
4. *Chronic schistosomiasis:* Praziquantel is the treatment of choice. Microbiological and ultrasound monitoring is recommended, especially if signs of organ involvement persist (strong recommendation; moderate level of evidence).
5. *Soil-transmitted helminth infections:* Treated with a single 400 mg dose of albendazole. Follow-up is only necessary if symptoms or eosinophilia persist. For specific infections like *Trichuris trichiura*, parasitological control is advised 3–4 weeks after treatment (strong recommendation; moderate level of evidence).
6. *Infection by Mycobacterium tuberculosis:* Treatment is recommended for all patients with no prior therapy. Active disease must be excluded, and treatment regimens should be tailored to individual risk. Follow-up should include regular clinical, biochemical, and hematological monitoring to assess toxicity, with adjustments based on comorbidities and concomitant treatments (strong recommendation; moderate level of evidence).

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Conflicts of interest

None declared.

Appendix A. Supplementary data

Supplementary data associated with this article can be found in the online version available at <https://doi.org/10.1016/j.eimce.2025.07.005>.